

SUMMER INTERNSHIP
CYBERSECURITY BLUE TEAM
Digital Forensics & Incident Response

Design and Implementation of a Lightweight DFIR and Threat Hunting Framework for Endpoint Investigation

1 Context & Motivation

Cyber threats are growing in both volume and sophistication, and organizations increasingly need professionals who can not only detect security incidents but also investigate them and respond effectively. Detection on its own is no longer enough; the ability to reconstruct what happened, understand attacker behavior, and contain the impact has become a core competency for modern security teams.

2 Internship Subject

Design and Implementation of a Lightweight DFIR and Threat Hunting Framework for Endpoint Investigation

The intern will design and build a lightweight, automated framework capable of collecting endpoint artifacts, detecting indicators of compromise, mapping observed behavior to attacker techniques, and producing structured investigation reports. The work pairs hands-on tool building with a genuine investigative methodology, so the outcome is both a working prototype and a repeatable process.

3 Objectives

Build a lightweight, automated framework that collects and analyzes endpoint artifacts on Windows and Linux systems.

- Detect suspicious behavior and indicators of compromise (IOCs) using YARA rules and Sigma detection rules.
- Map identified threats to the MITRE ATT&CK framework to produce structured, actionable intelligence.
- Automate the generation of clear investigation reports for incident response.

Simulate realistic attack scenarios to validate the framework's detection capabilities.

- Develop proactive threat hunting methods that transfer directly to a professional environment.

4 Scope of Work

4.1 Artifact Collection Module

A collection engine, written in Python, that extracts key endpoint artifacts:

- Running processes and their metadata
- Active network connections
- Scheduled tasks and startup programs
- Registry persistence keys (Windows)
- Security event logs and system logs

4.2 Detection & Analysis Engine

- Integration of YARA rules for malware and suspicious-file detection
- Implementation of Sigma rules for behavioral detection across log sources
- Correlation of collected artifacts against known IOC sources

4.3 MITRE ATT&CK Mapping

- Automatic tagging of detected behavior with MITRE ATT&CK tactics, techniques, and procedures
- Reconstruction and visualization of the attack chain based on identified techniques

4.4 Reporting & Dashboard

- Automated generation of structured investigation reports
- A summary view presenting key findings, risk levels, and recommended actions

5 Indicative Timeline

Phase	Description
Phase 1: Foundations	Study Windows and Linux forensic artifacts, configure Sysmon, and establish a baseline of normal system behavior.
Phase 2: Artifact Collection	Develop and test the automated collection engine for logs, processes, and persistence data.
Phase 3: Detection Engineering	Integrate YARA and Sigma rules, implement IOC correlation, and add MITRE ATT&CK mapping.
Phase 4: Reporting & Validation	Build the reporting view, simulate attacks in the lab, and document the methodology and findings.

6 Tools & Technologies

- **Programming:** Python
- **Artifact collection:** Sysmon, a custom lightweight Python agent (Velociraptor concepts as reference)
- **Detection:** YARA rules, Sigma rules
- **Forensic methodology:** Windows Event Log analysis, KAPE concepts
- **Threat intelligence:** MITRE ATT&CK framework, public IOC sources
- **Reporting:** SQLite with a lightweight Flask or FastAPI interface

7 Expected Deliverables

- A functional DFIR and threat hunting framework prototype
- A documented YARA and Sigma ruleset tailored for endpoint investigation
- A MITRE ATT&CK-mapped incident report based on simulated attack scenarios
- A technical report describing the methodology, findings, and lessons learned
- A final presentation for the internship defense